

Detecting Depression Tendencies in Twitter Users

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Abstract

The project aims to classify Tweets suggestive of depression from Tweets that are not and then extend the analysis to identify depression tendencies in Twitter users. Depression is a serious mental illness that requires understanding and care, otherwise it may lead to life-threatening consequences such as self-harm and suicide. We want to explore how Twitter can be used for diagnosing depression among people. We plan to first create the dataset by scraping Tweets from multiple users who show depression tendencies and label the tweets accordingly. We will then use multiple deep learning models such as RNN, LSTM, GRU, etc and predict the type of Tweet. Later, we aim to identify the performance of the different state-of-the-art models against the traditional models for this particular task, evaluate accuracy and then find out our best performing model. We can use the classifiers on any Twitter user and quantify the depression level for that particular user by analysing the Tweets. We expect that our findings would be useful for the Health Sciences Department who can predict depression among people and provide them preemptive help.

1 Introduction

In this project, we aim to tackle the classical NLP task of classification. Specifically, we want to classify Tweets as either "suggestive of depression" or "not suggestive of depression". A natural extension of our project is that given a Twitter user handle, determine the depression tendencies of the user. Depression is a mental illness that many people conceal and are unaware that it is affecting them. A lot of times, you can analyze a depressed person by their social activity. We chose Twitter data for this analysis because Twitter is a very popular platform for sharing information in real-time. The brevity of the Tweets allow users

to quickly share events and sometimes their emotions with their followers. Also, a lot of user profiles are public unlike Facebook or Snapchat. Detecting depression through NLP is a more convoluted task than simple sentiment analysis, since the labels "not depressed/ depressed" cannot be equated with the labels "happy/sad". The indications of the depression in tweets are often subtle, thus not immediately obvious to the human reader.

In the study "CLPsych 2015 Shared Task: Depression and PTSD on Twitter" ([URL](#)) by Coppersmith et al. several approaches were proposed to identify whether a Twitter user suffered from depression or not. Resnik et al., 2015 used topic modelling for this classification task, while Preotiuc-Pietro et al., 2015 examined a wide variety of methods for inferring topics automatically, combined with binary unigram vectors (i.e., "did this user ever use this word?"), and scored using straightforward regression methods. Pedersen, 2015 took a rule-based approach to these tasks, and as such provides a point to examine how powerful simple, raw language features are in this context.

While these approaches used a good dataset that included Twitter users that were clinically classified as depressed, they did not include the state-of-the-art NLP techniques and relied on raw language features and due to privacy concerns, did not release the dataset.

We propose to include a larger dataset as opposed to the one used by earlier researchers. We would also train state-of-the-art NLP models on this classification task. However, a direct comparison with the earlier approaches is not possible due to the unavailability of the dataset which they had used.

To evaluate the performance of our models, we plan to do both a quantitative analysis as well as a qualitative analysis. We also have decided to test this classification algorithm over a twitter user. Using our models, we will be able to decide depression levels for a particular user.

1. Qualitative Analysis

- (a) Comparison of different models like RNN, CNN, LSTM, GRU on word embeddings. We have planned to highlight the differences between the performances across models and mention the potential cause/reasoning for these differences.
- (b) Compare performance between word-based and character-based embeddings for different models- We expect that for character-based embeddings, the training time will increase but it might perform better for some examples.

It will be a good idea to even point out some relevant examples from training data that were incorrectly classified by one model and correctly classified for other OR examples that were incorrectly classified by word embeddings but correctly classified by character embeddings.

2. Quantitative Analysis

- (a) Evaluate the accuracy, precision, recall, F1 scores across different models
- (b) Measure training and validation accuracy for each model.
- (c) Measure training and validation loss for each model.

3. Depression Levels for a particular user

We now plan to use the best model after computing the different metrics and apply that in finding out the depression level for one user. We plan to quantify this number to indicate how depressed that individual is. This information will reveal some insights which can then be used by health sciences for further diagnosis of that individual.

2 Current Progress

The baseline model we are considering for our task is Logistic Regression. We use the Logistic

Regression defined in scikit library. In order to apply the logistic regression, we decided to calculate the sentence representations for this model from word vectors. We calculated sentence representations in 2 different ways, Firstly, we use tfidf to compute the sentence representation. The size of this representation is equivalent to the size of vocabulary of the dataset. The sentence representation is created by different word frequency and ends up having a lot of sparse values. Secondly, we generate sentence representation from the average of the word vectors. For each word we computed the word representation from word embeddings which are generated from the Word2Vec function from the genism python library. Lastly we will try the model on different hyperparameters and solvers for optimal results.

We have currently identified the dataset by web page scrapping and identifying labels to those data. We have identified the challenges while pushing the dataset to our baseline model such as removing the sentences with less than 5 words, restricting the word limits, removing the emojis and hash tags. We have removed the tweets which contained media format. For our baseline model, currently we are identifying the hyper paramaters and the ranges for those parameters for optimal result. The evaluation metrics for the system is decided and the baseline model will be evaulted on it for our threshold results.

3 Dataset

To train any NLP model for the classification task, we need a substantial training dataset. For the task of detecting depression in Tweets, we need Tweets that are suggestive of depression along with Tweets that are not. But, it is difficult to find Tweets that are linked with depression mainly due to the fact that any Tweet containing keywords such as depression, sad might not be directly linked to depression. A number of Tweets containing such keywords are actually aimed at helping people suffering from depression. Therefore, to find Tweets indicative of depression, we tried to find public accounts that showed signs of depression and generally Tweets content indicative of depression. We were able to find around 20k such Tweets, but we would have to do further analysis to remove Tweets that are shorter than 4-5 words and contain images or videos. We would

also need to add positive/neutral Tweets that are not suggestive of Depression. This is an easier task as we can get many such Tweets from different public accounts on Twitter such as News channels, celebrity accounts, accounts posting motivating/positive quotes etc.

4 Methodology

Besides the baseline model defined below, we would like to evaluate the performance of popular models such as -

- DAN
- LSTM
- GRU

We will also evaluate some state-of-the-art models on this task such as -

- BERT (Bidirectional Encoder Representations from Transformers)
- XLNet

References

- Deep Learning for Depression Detection of Twitter Users - <https://www.aclweb.org/anthology/W15-1204>
- CLPsych 2015 Shared Task: Depression and PTSD on Twitter - <https://www.aclweb.org/anthology/W18-0609>